

Final Exam Review

Course: STAT451

Classes 24A01, 24A02

Part I. Probability (1 question)

- Computing probabilities related to the binomial distribution and the normal distribution.
- The Central Limit Theorem; application to approximating the binomial distribution by the normal distribution.

Part II. Statistics (4 questions)

1. Descriptive Statistics

Computing sample statistics: mean, variance, standard deviation, and median.

2. Confidence Intervals

- (a) Confidence interval for the population mean (known and unknown variance; large samples, small samples).
- (b) Confidence interval for a proportion.
- (c) Margin of error; determining the sample size.

3. Hypothesis Testing - One-Sample Case

- (a) Hypothesis testing for the population mean (known and unknown variance; large samples, small samples).
- (b) Hypothesis testing for a proportion.
- (c) Computing the p -value (for cases using the z -statistic).

4. Hypothesis Testing - Two Independent Samples

- (a) Comparing two population means (known and unknown variance; large samples, small samples with equal/unequal variances).
- (b) Comparing two proportions.
- (c) Computing the p -value (for cases using the z -statistic).
- (d) Confidence interval for the difference of two means and for the difference of two proportions (open-ended question).

5. Two Dependent Samples

The paired t -test.

6. Simple Linear Regression

- (a) Determining the regression line; interpretation of the regression coefficients.
- (b) Assessing the model: the coefficient of determination (R^2).
- (c) The sample correlation coefficient and its interpretation.
- (d) Making predictions from the estimated regression line.

Exam Format

1. **Duration:** 90 minutes.
2. **Permitted materials:** Students are allowed to bring **only one** A4-sized sheet of paper (both sides) containing necessary formulas. This sheet must be either typed or photocopied. If the formulas are handwritten, the sheet must still be photocopied before being brought into the examination room (handwritten formula sheets will otherwise be treated as unauthorized materials). The formula sheet must clearly display the student's **full name** and **student ID**.